Validation of Network-Flow Feature Engineering for Explainable Intrusion Detection

A NovaShield Case Study

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> **Research finding, in one sentence:** When the current NovaShield feature-processing implementation was re-run against five preserved synthetic packet captures and their CICFlowMeter reference outputs, **all 468 numerical comparisons matched** under the test comparator's rules, and **all 10 source-evidence hash checks passed**.

Important boundary: These are *feature-agreement* results for selected synthetic tests, **not** a claim that NovaShield detects 100% of attacks, matches CICFlowMeter on all traffic, or is ready for operational deployment.

Abstract

Machine-learning intrusion detection relies on measurements extracted from network traffic. If packet grouping, timing, or traffic-rate calculations are inconsistent, two systems can interpret the same network activity differently—even when they use the same classification model. This report investigates the agreement between NovaShield's custom Python network-flow feature pipeline and reference outputs previously produced by CICFlowMeter. Five preserved synthetic packet-capture fixtures were analysed using NovaShield's current flow assembler, feature exporter, and comparison functions. The original experiment records contained a flow-count mismatch in Test 5 and numerical differences in Tests 7 and 8. A subsequent read-only local revalidation of the unchanged packet captures and reference CSV files produced **five PASS results: 156 matching numerical comparisons in Test 4 and 78 in each of Tests 5, 7, 8, and 9 (468/468 in total)**, with zero reported differences. A separate SHA-256 audit verified the integrity of all five packet captures and five reference CSV files against the hashes recorded in the original experiments. The result supports agreement for the selected features and fixtures under the implemented comparison logic. It does not establish model detection accuracy, broad feature equivalence, or independent replication. The case study demonstrates why validating measurements is a necessary foundation for explainable security analytics.

Keywords: intrusion detection; network-flow features; CICFlowMeter; explainable AI; software verification; reproducibility; cybersecurity research

1. Why this research matters

A network intrusion detection system (NIDS) attempts to recognise suspicious behaviour in network activity. Machine-learning versions generally do not make predictions directly from a human-readable description of a connection. Instead, they receive a set of numbers describing that activity—for example, how many packets were exchanged, how long a connection lasted, and how quickly data moved.

These numbers are called **network-flow features**. They are the measurements the AI is asked to trust.

A simple analogy: Imagine two doctors measuring the same patient's temperature with different thermometers. Before comparing their diagnoses, we would want to know whether both thermometers report consistent measurements. In much the same way, before trusting a network-security model, we should examine whether its traffic measurements are consistent with an established reference process.

This question also matters for *explainable AI*. Techniques such as SHAP can describe which model inputs influenced a prediction, but an explanation is only as meaningful as the input measurements it refers to. A persuasive explanation cannot compensate for an incorrect traffic feature.

NovaShield is an independent cybersecurity research prototype exploring network-flow observation and feature engineering (OPTIC), offline machine-learning and explanation (CORTEX), and analyst-oriented SOC workflows. **This report evaluates feature engineering only.** It does not test the complete platform or its response mechanisms.

2. Research question and contribution

Research question: *For five preserved, controlled synthetic network-traffic fixtures, do features generated by the current NovaShield pipeline agree with the stored CICFlowMeter reference outputs under NovaShield's comparison procedure?*

The contribution is a bounded, traceable engineering result rather than a new claim about classifier superiority:

1. **A documented baseline:** original stored results identify an earlier flow-count disagreement and feature differences.
2. **A corrected implementation recheck:** the current pipeline passes all five selected fixtures against the same stored reference data.
3. **An integrity check:** the ten PCAP/CSV files used for revalidation match their previously recorded SHA-256 hashes.
4. **Transparent limitations:** the study identifies what these comparisons can—and cannot—establish.

3. What was tested

3.1 The data

The experiment used five existing **synthetic packet-capture (PCAP) fixtures**, identified as Tests **4, 5, 7, 8, and 9**. These fixtures were generated for controlled feature-calculation experiments; they are not a representative sample of live organisational networks. The successful comparison involved **six network flows** in total: two from Test 4 and one from each other selected test.

For every fixture, the local research directory retains four types of evidence:

Evidence	Purpose
`test_N.pcap`	Original synthetic packets used as input
`test_N.pcap_Flow.csv`	Preserved CICFlowMeter reference feature output
`comparison.json`	Earlier comparison outcome and recorded file hashes
`cicflowmeter_log.txt`	Reference-tool execution log

Here, `N` stands for the fixture number. The existing `.gitignore` excludes PCAP and CSV files from ordinary Git staging; therefore, **the public GitHub repository does not yet constitute a complete independent reproduction package**.

3.2 The software

The local revalidation used three project components:

- **`scripts/novashield_flow_assembler.py`** — groups packets into flows using `FlowAssembler`.
- **`scripts/novashield_feature_exporter.py`** — calculates the numerical feature representation using `export_partial_features`.
- **`research_fixtures/automated_validation/run_tests.py`** — compares NovaShield features with the preserved reference rows using its `compare` logic.

CICFlowMeter was **not rerun** during the latest check. The comparison used its **previously saved reference CSVs**. Consequently, this is a revalidation of NovaShield against a preserved reference—not an independent audit of CICFlowMeter itself.

4. Method: from packets to comparisons

The revalidation was intentionally read-only. It did not overwrite the original PCAPs, reference CSVs, saved JSON comparisons, or logs.

The validation process

1. **Read a saved packet capture** with Scapy's `rdpcap`.
2. **Rebuild network flows** by passing each packet into the current `FlowAssembler`.
3. **Apply the end-of-capture rule** used by the test runner: include completed flows and retain unfinished active flows only when they contain more than one packet.
4. **Calculate features** using `export_partial_features`.

5. **Load the reference CSV** with Python's `csv.DictReader`.
6. **Compare the generated and reference flow rows** using the project's existing comparison functions.
7. **Read the outcome**: matched numerical values, differences, flow counts, and PASS/REVIEW status.

The comparator uses its defined feature mappings, numerical tolerances, and flow-pairing rules. Accordingly, **"matching" means matching under that implementation's criteria**; it does not mean every floating-point value is bit-for-bit identical. The precise threshold values and mapping rules still need to be transcribed into a versioned methodology appendix before external submission.

Separately, SHA-256 hashes were recalculated for the original five PCAPs and five CICFlowMeter CSVs. Each was checked against the value already stored in that fixture's original `comparison.json` file. This establishes that the files used in revalidation are consistent with those saved hashes; it is **not** a third-party chain-of-custody certification.

5. Results

5.1 What the original experiments showed

The saved historical JSON comparisons are retained as evidence of the earlier development state.

Fixture	Original saved status	Earlier observation
Test 4	**PASS**	156 matches, no differences; two flows in both implementations
Test 5	**FLOW_COUNT_MISMATCH**	Two NovaShield flows versus one reference flow
Test 7	**REVIEW**	77 matches; one numerical difference
Test 8	**REVIEW**	76 matches; two numerical differences
Test 9	**PASS**	78 matches, no differences

These earlier results are **not deleted, overwritten, or retrospectively labelled PASS**. Retaining them makes the development process more transparent.

5.2 What the current code produced

The subsequent local revalidation used the *same preserved input PCAPs and reference CSVs*, but the **current NovaShield implementation**.

Fixture	Current status	Matches	Differences	NovaShield flows	Reference flows
Test 4	**PASS**	156	0	2	2

Fixture	Current status	Matches	Differences	NovaShield flows	Reference flows
Test 5	**PASS**	78	0	1	1
Test 7	**PASS**	78	0	1	1
Test 8	**PASS**	78	0	1	1
Test 9	**PASS**	78	0	1	1
Total	**5 / 5 PASS**	**468**	**0**	**6**	**6**

Primary finding: 468 of 468 numerical comparisons matched; zero differences were reported under the comparator rules. All five selected tests returned `PASS`.

This is a *feature validation rate for this test set*, not a classification accuracy statistic. No attacks were classified in this revalidation, so it would be incorrect to call this "100% attack detection accuracy."

5.3 Evidence integrity

A further read-only check compared each evidence file's current SHA-256 hash with the value in the original comparison record.

Integrity check	Outcome
Five original PCAP hashes	**5 / 5 verified**
Five CICFlowMeter CSV hashes	**5 / 5 verified**
Overall	**10 / 10 verified**

The integrity result strengthens confidence that the local recheck used the intended preserved files. It does not independently establish the provenance or correctness of the reference tool's output.

6. What changed—and what we can reasonably conclude

The comparison history shows a meaningful engineering progression. An earlier implementation did not agree with the reference on some selected cases; the current implementation does. Prior development investigations identified **flow lifecycle handling**, including the treatment of unfinished single-packet flows, and **duration quantisation in rate calculations** as relevant technical issues.

However, this report does **not** claim that each original mismatch has been conclusively attributed to a particular code change. That stronger causal claim would require a documented commit comparison and dedicated regression test for each defect.

What the evidence supports: for the five selected synthetic fixtures, the present NovaShield pipeline generates numerical outputs that agree with the preserved CICFlowMeter reference outputs under the implemented comparison criteria.

Why this matters: a machine-learning security model depends on its input features. Establishing controlled consistency of those features is one necessary step toward meaningful model evaluation and explanation. It is not, by itself, proof that the full detection system is reliable.

7. Limitations and research integrity

This study is intentionally modest in scope. Its principal limitations are:

- **Limited coverage.** Five synthetic fixtures and six flows do not cover normal or malicious traffic diversity, operating-system differences, packet loss, fragmented traffic, or deployment conditions.
- **Comparator dependence.** The numerical results reflect the project's current feature mapping, tolerance, and pairing implementation; their exact rules need to be documented for external scrutiny.
- **Reference rather than ground truth.** Agreement with stored CICFlowMeter output is useful, but CICFlowMeter is not an infallible oracle.
- **Local revalidation.** The successful rerun was conducted in the project's local environment, not independently reproduced by a third-party researcher.
- **Incomplete public artefacts.** The public repository includes the comparison scripts and historical JSON results, but the PCAP/CSV reference inputs are presently excluded from Git and have not been packaged for open reproduction.
- **Narrow protocol support.** An additional attempted fixture produced an IPv4/TCP-only support error in the tested processing path. That fixture is **not** part of the 468/468 result.
- **No detection-performance claims.** This experiment does not measure true positive rate, false positive rate, precision, recall, F1 score, calibration, speed, or practical protection against attacks.
- **No verified causal attribution yet.** The historical differences and the current passes establish a change in outcome, but more focused regression evidence is needed to isolate each correction.

These boundaries are not weaknesses to conceal. They define the strength of the claim and show where the next experiments should begin.

8. Reproducibility: what is complete and what comes next

The local evidence has been preserved and checked, but **external reproduction is still a work in progress**. Before treating the report as a final archival research publication, the next version should include:

1. A standalone **offline verification script** that reads the selected fixtures and saved CICFlowMeter CSVs without using SSH, sudo, or a live network interface.
2. A **new, dated JSON results record** for the successful 468/468 run, leaving all historical `comparison.json` and `summary.json` files unchanged.
3. The exact **Git commit ID**, Python and Scapy versions, CICFlowMeter reference version, comparator mapping, and numerical tolerances.
4. Carefully reviewed, sharable **synthetic PCAP and reference CSV files**, or a documented, deterministic regeneration procedure with hash values.

5. A clear table connecting each **earlier discrepancy** to a verified regression test and source-code correction.
6. Reproduction in a clean environment by another investigator, where feasible.

A future broader study should test additional protocol behaviours, traffic patterns, boundary cases, time quantisation effects, cross-dataset compatibility, and implications for downstream intrusion-detection predictions and SHAP explanations.

9. Conclusion

This case study addresses a practical but often overlooked research question: **can an AI security system trust the measurements it uses?**

Using five preserved synthetic network-capture fixtures, the current NovaShield network-flow pipeline passed all five local comparisons against stored CICFlowMeter outputs: **468 out of 468 numerical comparisons matched**, and no differences were reported. A separate SHA-256 audit verified all ten original PCAP and reference CSV files against their recorded hashes. Earlier mismatch records were preserved rather than overwritten, providing an honest record of the work's progression.

The result is an encouraging, bounded validation of feature-engineering consistency. Its value lies in the careful treatment of measurement correctness, evidence preservation, and explicit research limitations—not in claiming a finished or infallible intrusion-detection system. The next scientific priority is independent reproducibility across a broader and more representative set of network traffic.

References and research record

[1] Batool, M. (2026). *NovaShield: Explainable AI Cybersecurity Research Prototype* (Version 0.1.0) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.24056070

[2] NovaShield public source repository. https://github.com/mariabatool869-star/nova_shield

[3] NovaShield validation source: `research_fixtures/automated_validation/run_tests.py`, and flow/feature processing in the repository's `scripts/` directory.

[4] CICFlowMeter, reference flow-feature extraction software. The precise build/configuration used to produce the saved CSVs must be documented before final external publication.

Citation and publication note

This is a technical prepublication manuscript, not a peer-reviewed article. The DOI above identifies the **NovaShield v0.1.0 software archive**, not this report. The experimental outcomes are reported from local terminal-run verification and preserved test files; an independently published replication package and a separate report DOI do not yet exist. The manuscript may be revised before deposit as a standalone research output.